Monitoring and Logging System using Prometheus & Grafana

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Technologies: Docker | Prometheus | Grafana | AWS EC2

PROJECT OVERVIEW

Objective

To monitor server performance and visualize real-time metrics using Prometheus and Grafana.

Tools & Technologies

- Prometheus (Metrics Collection)
- Grafana (Visualization)
- Node Exporter (System Metrics)
- Docker (Containerization)
- AWS EC2 (Hosting Server)

Project Description

This project demonstrates the implementation of a real-time monitoring system using Prometheus and Grafana. Metrics from the server are collected using Node Exporter and visualized through Grafana dashboards, enabling effective system monitoring.

IMPLEMENTATION

Steps Performed

1. EC2 Setup

- Created EC2 instance (Amazon Linux)
- Configured security group with ports:
    - 22 (SSH)
    - 3000 (Grafana)
    - 9090 (Prometheus)
    - 9100 (Node Exporter)

2. Docker Setup

Installed and configured Docker:

```
sudo yum install docker -y
sudo systemctl start docker
sudo systemctl enable docker
```

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3. Monitoring Setup

- Created Prometheus configuration file
- Ran Node Exporter container
- Ran Prometheus container
- Verified target status (UP)

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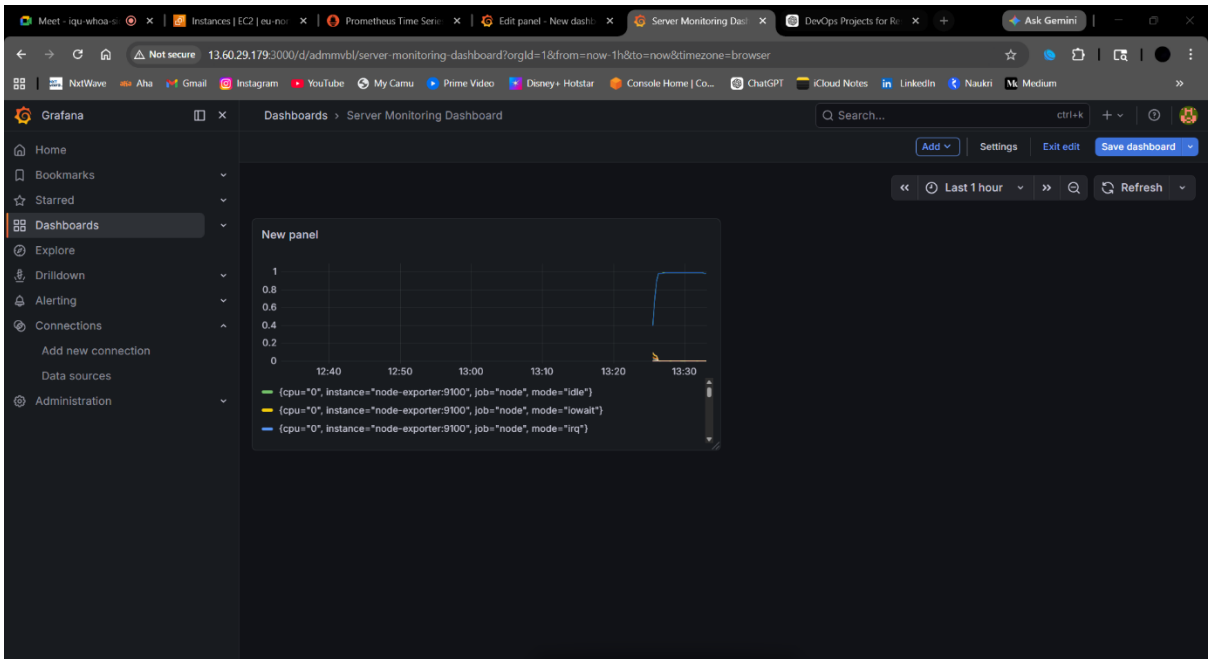
4. Grafana Setup

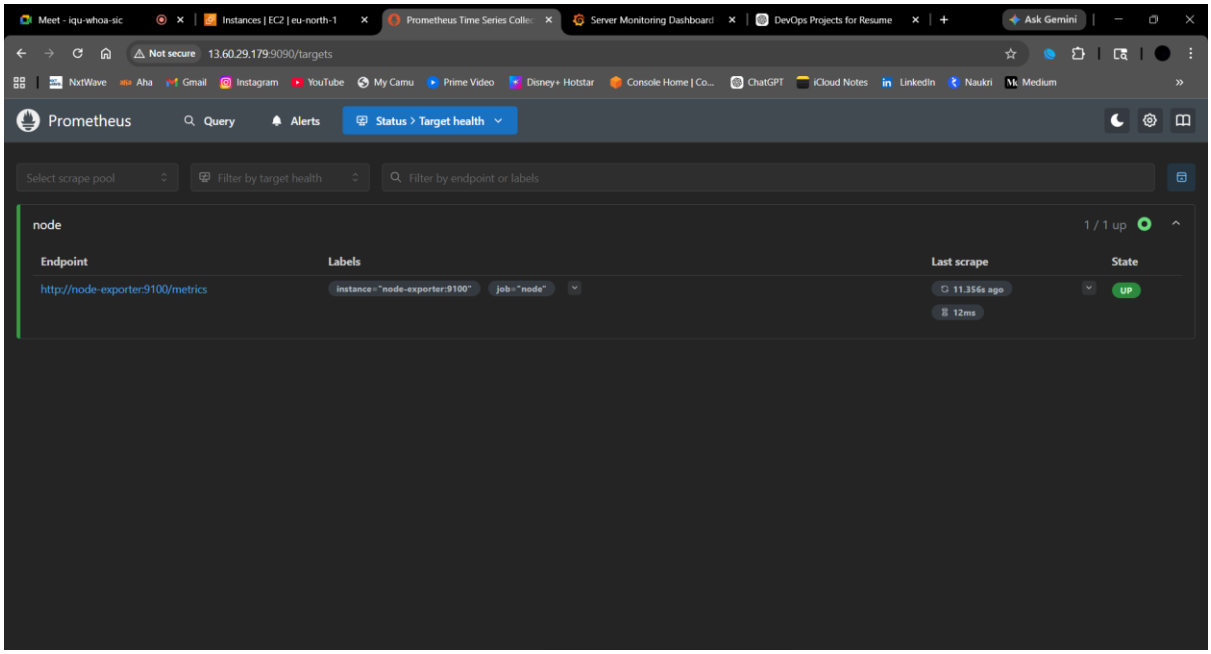
- Accessed Grafana web UI
- Added Prometheus as data source
- Created dashboard
- Visualized CPU metrics

Grafana Dashboard



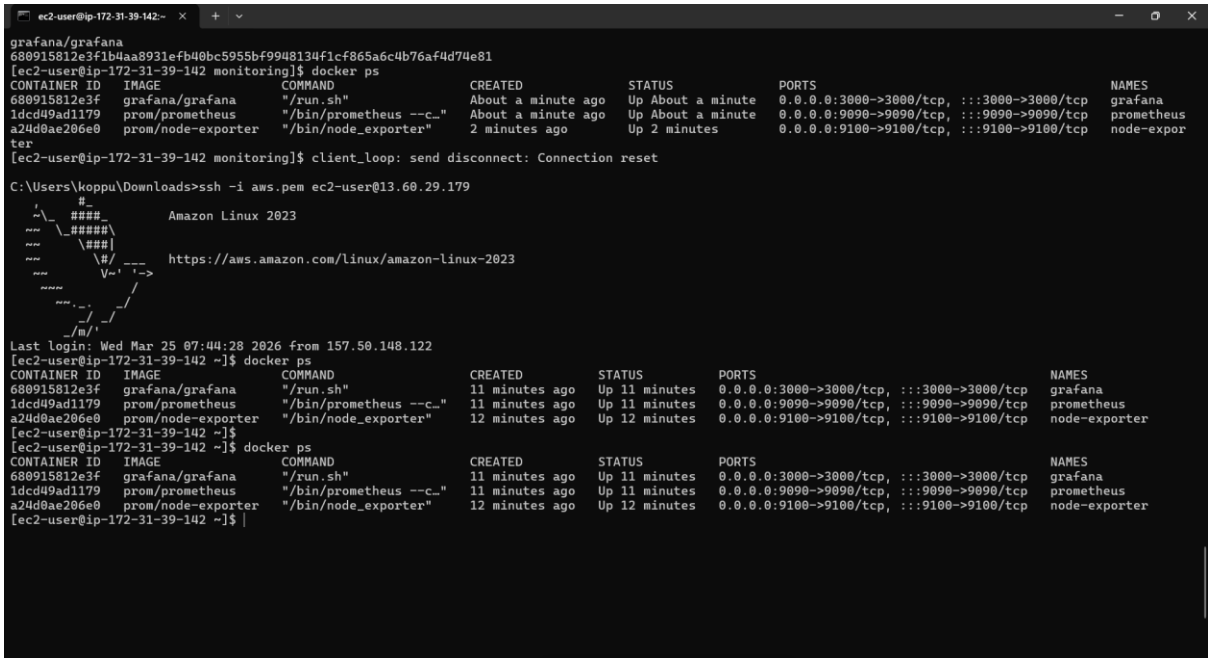
Grafana dashboard displaying real-time CPU metrics.

Prometheus Targets



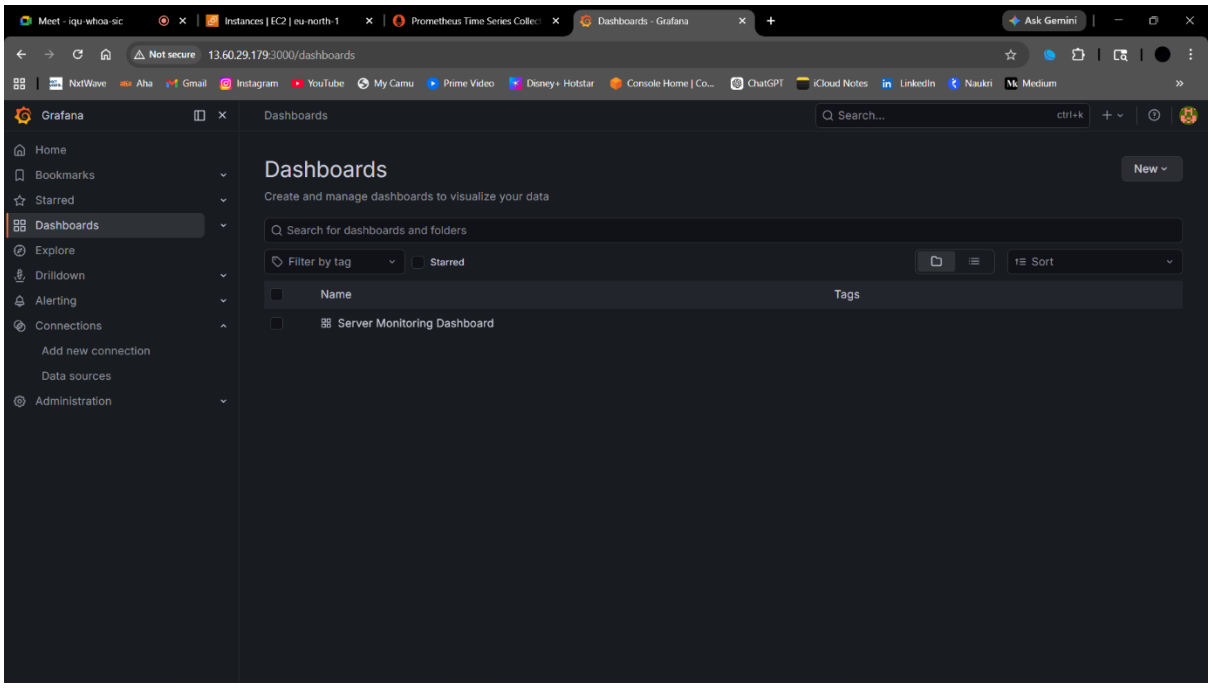
Prometheus successfully scraping metrics from Node Exporter (Target Status: UP).

Docker Containers



Prometheus, Grafana, and Node Exporter containers running successfully.

Dashboard List



Monitoring dashboard successfully created and visible in Grafana.

OUTPUT & CONCLUSION

Output

- Real-time monitoring system implemented
- CPU usage visualized in Grafana
- Prometheus integrated successfully
- Docker-based monitoring setup completed

Conclusion

This project demonstrates the implementation of a monitoring system using Prometheus and Grafana, enabling real-time observability of system performance and metrics.